

Evaluating the Performance of the Finetuned Quantized Llama 3 based on relevance, truthfulness, naturalness, and safety

Notebook for the Eloquent Lab at CLEF 2025

Rohit Gunti^{1,†} and Abebe Rorissa^{2,*,†}

¹ University of Tennessee, Knoxville, School of Information Sciences, 1345 Circle Park Drive, Suite 412, TN, USA

² University of Tennessee, Knoxville, School of Information Sciences, 1345 Circle Park Drive, Suite 451, TN, USA

Abstract

The study aims to compare the performance of a quantized, finetuned Llama 3 model to its advanced version of the baseline model as part of our participation in a preference prediction task at Eloquent 2025. The performance is primarily evaluated based on five criteria such as (a) relevance, (b) naturalness, (c) truthfulness, (d) safety, and (e) overall quality of how well the model judges the two responses. Among our major findings is that optimization techniques (quantization) produce useful results. Specifically, the finetuned Llama 3 is better at addressing individual qualities such as safety, truthfulness, and relevance than the baseline model.

Keywords

Naturalness, Truthfulness, Safety, Overall Quality, Quantization, Llama 3, Llama 3.1, BERT, ROUGE

1. Introduction

Prior works submitted to the Eloquent Lab 2024 primarily focused on evaluating the quality of system responses and task reports [1]. The authors of those submissions addressed the training costs, configuration, and resources utilized during evaluations [2, 3, 4, 5, 6, 7]. Little attention is given in the extant literature to system optimization and reproducibility for interdisciplinary scholars. Only a few studies avoid expensive processing to keep the system computationally light [8]. The study by [8] utilizes relatively small 7B parameters and few-shot inference (no fine-tuning) and even employs a quantized version of the LLM (Mistral). Moreover, the focus is on a single task of detecting hallucinations in LLMs, specifically leveraging prompt engineering techniques for this purpose.

Organizations have consistently emphasized the importance of optimization. Even before the AI surge, some non-profit organizations, such as libraries, notably used optimization techniques to analyze how applications ran in real-time and adjust their structure for better performance [9]. Similarly, other studies focused on practical application and used ANN optimization techniques to enhance system performance [10]. There is some evidence that ML optimization techniques can lead to performance improvement [11]. Considering the billions of parameters an LLM is trained on, it is essential to reduce the computational load of LLMs for some organizations with tailored budgets, allowing them to fine-tune it for multiple tasks before deployment. There is ongoing exploration on training the LLM with optimization techniques [12, 13, 14]. Therefore, this study contributes to the ongoing exploratory efforts by participating in Eloquent 2025 and reporting the optimized methodology and findings of Unsloth finetuned Llama 3. In this paper, we share our findings on preference prediction assessments monitored by Eloquent 2025.

^{1*} Corresponding author.

✉ rgunti@vols.utk.edu (R. Gunti);  0000-0002-5239-2419 (R. Gunti); 0000-0002-5300-617X (A. Rorissa)

2. Methodology

The following three sections describe our methods: (1) Data Collection, (2) Data Preprocessing, and (3) Finetuning. Lastly, the findings section includes the training results and shared results evaluated by the Preference Prediction task (2025) committee.

2.1. Data

The dataset (2025 validation data) used for finetuning the Llama 3 model is provided by the preference prediction committee. The dataset contains 99 JSON items where each item is similar to alpaca format but has more fields in addition to instruction, input, and output. This study uses this dataset for finetuning to make an LLM generate better response to a use instruction based on multiple quality criteria such as relevance, truthfulness, naturalness, and safety. Hence, the validation dataset is referred to as the finetune dataset.

2.2. Finetune

Llava 3 using a Low Rank Adaptation (LoRa) approach was utilized for efficient memory usage. Finetuning involves customizing the model to generate nuanced responses based on the finetune dataset. Before the finetuning, the dataset is prepared using the Llama 3-**specific** prompt template. Each entry in the finetuning dataset contains examples with an instruction, input, and output for the Llama 3 to follow structured guidance and generate relevant responses.

2.3. Lora Configuration

The Llama 3 model, loaded in 4-bit quantization for efficiency, is set up using a specific training configuration. Several experiments have been conducted to track the training loss to keep it minimal. To supervise the finetuning, the SFT trainer is enabled. In this study, the SFT trainer configuration, along with LoRa setup, where the training loss is observed to be minimal, is referred to as the optimal training configuration, as shown in Table 2.

3. Findings

In our findings, we include two kinds of evaluation where each compares baseline scores and fine-tuned Llama 3 scores as shown in Tables 2 and 3. Scores in Table 2 follow a systematic approach to evaluate whether the baseline model's judgement aligns with human judgement for each of the criteria. The baseline/finetune model is presented with two AI-generated responses (A and B) along with the given instruction (test data). The test data has a total of 1,248 items. Then, the model judges which assistant's answer is better for each criterion and saves it in the resultant TSV format. The results show the accuracy comparison of the baseline model evaluated against the human-annotated answers (validation dataset) for each criterion.

The scores in the middle columns for baseline Llama 3.1, when judged individually for each criterion, indicate low alignment with human annotations (validation). However, the scores are notably higher for overall quality (42.42) than individual judgments, indicating that the baseline Llama 3.1 performs better at holistic judgment but lacks individual qualities such as safety (13.13) and

truthfulness (11.11). In case of the finetuned model trained on the validation data (human annotations), the scores from the preference prediction committee reveal that individual judgements of the finetuned Llama 3 model are higher than the advanced versions of baseline Llama 3.1. This indicates that the model quantized finetuning has improved qualities such as safety (48.96), relevance (39.98), and truthfulness (38.62), which are major concerns. However, the overall quality (33.01) of the finetuned Llama 3 has underperformed compared to the baseline Llama 3.1 model. This score indicates that the finetuned model lacks a balanced factor effectively when making an overall judgment. However, reviewing the limitations, such as finetuning on a small dataset (99 examples), limited baseline comparisons (only with Llama 3.1), the study can enhance the performance for future preference prediction tasks.

4. Conclusion

Our findings are competitive in the preference prediction 2025 task, ranking in second position (Team UTK). Since the study focuses on observing whether the quantized finetuned llama 3 provides better and useful results without compromising performance, further performance enhancement is out of the study’s scope. However, the training insights (configurations) and findings will serve as the evidence to explore the potential of quantized fine-tune models.

5. Tables

Table 1
Training, model and Lora configurations

Training Configurations (SFT Trainer)	
Batch Size	2
Gradient Accumulation Steps	4
Warm-up Steps	5
Max Steps	-1
Epochs	5
Learning Rate	2e-5
Weight Decay	0.01
Model Configurations	
Baseline	unsloth/llama-3-8b-Instruct
Maximum sequence length	2048
Quantization	4-bit
LoRa Configurations	
R value	16
alpha	16
Target modules	q_proj; k_proj; v_proj; o_proj; gate_proj; up_proj; down_proj

Table 2

The criteria, baseline scores, and finetuned model scores when evaluated for sub-task one on the preference prediction test dataset 2025.

Criteria	Llama 3.1 baseline Scores	Finetune Llama 3 Scores
Relevance	23.23	39.98
Naturalness	29.29	33.01
Truthfulness	11.11	38.62
Safety	13.13	48.96
Overall Quality	42.42	33.01

Table 3

The criteria, baseline scores, and finetuned model scores when evaluated for subtask 2 on the preference prediction test dataset 2025.

Criteria	Llama 3.1 baseline Scores	Finetune Llama 3 Scores
relevance_rougeL	9.643	10.95
relevance_bertscore_f1	83.747	84.5
naturalness_rougeL	8.769	8.54
naturalness_bertscore_f1	83.476	83.82
truthfulness_rougeL	8.553	9.02
truthfulness_bertscore_f1	82.167	82.73
safety_rougeL	5.767	6.02
safety_bertscore_f1	81.739	82.10
overall_quality_rougeL	9.164	10.45
overall_quality_bertscore_f1	83.537	84.2

Acknowledgements

We would like to thank the University of Tennessee, Knoxville’s High Performance & Scientific Computing Team for providing us with access to the Nvidia H100 GPU for finetuning and evaluating the Llama 3 and Llama 3.1 models

Declaration on Generative AI

During the preparation of this work, the authors used Grammarly and Copilot in order to: Grammar and spelling check. After using these tool(s)/service(s), the authors reviewed and edited the content as needed and take full responsibility for the publication’s content.

References

- [1] Jussi Karlgren, Luise Dürlich, Evangelia Gogoulou, Liane Guillou, Joakim Nivre, Magnus Sahlgren, Aarne Talman, and Shorouq Zahra. 2024. Overview of ELOQUENT 2024—Shared Tasks for Evaluating Generative Language Model Quality. In *Experimental IR Meets Multilinguality, Multimodality, and Interaction: 15th International Conference of the CLEF Association, CLEF 2024, Grenoble, France, September 9–12, 2024, Proceedings, Part II*. Springer-Verlag, Berlin, Heidelberg, 53–72. https://doi-org.utk.idm.oclc.org/10.1007/978-3-031-71908-0_3
- [2] Karlgren, J., & Talman, A. (2024). ELOQUENT 2024-Topical Quiz Task. In *Conference and Labs of the Evaluation Forum* (pp. 687-690). CEUR-WS. org.
- [3] Dürlich, L., Gogoulou, E., Guillou, L., Nivre, J., & Zahra, S. (2024). Overview of the CLEF-2024 Eloquent Lab: Task 2 on HalluciGen. In *25th Working Notes of the Conference and Labs of the Evaluation Forum, CLEF 2024. Grenoble. 9 September 2024 through 12 September 2024* (Vol. 3740, pp. 691-702). CEUR-WS.
- [4] Sahlgren, M., Karlgren, J. J., Dürlich, L., Gogoulou, E., Talman, A., & Zahra, S. (2024). ELOQUENT 2024—Robustness Task. In *Conference and Labs of the Evaluation Forum* (pp. 703-707). CEUR-WS. org.
- [5] Neralla, V., & de Vroe, S. B. (2024). Evaluating Poro-34B-chat and mistral-7B-instruct-v0. 1: LLM system description for ELOQUENT at CLEF 2024. *Working Notes of CLEF*.
- [6] Sahlgren, M., Karlgren, J. J., Dürlich, L., Gogoulou, E., Talman, A., & Zahra, S. (2024). ELOQUENT 2024—Robustness Task. In *Conference and Labs of the Evaluation Forum* (pp. 703-707). CEUR-WS. org.
- [7] Bui, A. T. M., Brech, S. F., Hußfeldt, N., Jennert, T., Ullrich, M., Breuer, T., ... & Schaer, P. (2024). The Two Sides of the Coin: Hallucination Generation and Detection with LLMs as Evaluators for LLMs. *arXiv preprint arXiv:2407.09152*.
- [8] Siino, M., & Tinnirello, I. (2024). GPT hallucination detection through prompt engineering. *Working Notes of CLEF*.
- [9] Kistler, T., & Franz, M. (2003). Continuous program optimization: A case study. *ACM Transactions on Programming Languages and Systems (TOPLAS)*, 25(4), 500-548.
- [10] Fei, N. C., Mehat, N. M., & Kamaruddin, S. (2013). Practical applications of Taguchi method for optimization of processing parameters for plastic injection moulding: a retrospective review. *International Scholarly Research Notices*, 2013(1), 462174.
- [11] Arkabaev, N., Rahimov, E., Abdullaev, A., Padmanaban, H., & Salmanov, V. (2025). Modelling and analysis of optimization algorithms. *Jurnal Ilmiah Ilmu Terapan Universitas Jambi*, 9(1), 161-177.
- [12] Wang, Y., Afzal, M. M., Li, Z., Zhou, J., Feng, C., Guo, S., & Quek, T. Q. (2025). Large Language Model as a Catalyst: A Paradigm Shift in Base Station Siting Optimization. *IEEE Transactions on Cognitive Communications and Networking*.
- [13] Huang, Y., Wu, S., Zhang, W., Wu, J., Feng, L., & Tan, K. C. (2025). Autonomous multi-objective optimization using large language model. *IEEE Transactions on Evolutionary Computation*.
- [14] Huang, C., Tang, Z., Hu, S., Jiang, R., Zheng, X., Ge, D., ... & Wang, Z. (2025). Orlm: A customizable framework in training large models for automated optimization modeling. *Operations Research*.